

Optimisation Problems

Bird & de Moor, *Algebra of Programming* ch. 7 —
in Freyd diagram order

Conventions

Composition is Juxtaposition in diagram order: $RS =$ “first R , then S ”. Objects α, β, γ ; relations Q, R, S, T, U, V, X, Y ; **maps** (entire and simple) f, g, h, k, m ; coreflexives $C \subset 1$. R° converse, $\subset$ containment, $\cap, \cup$ meet/join, $-$ subtraction, $\emptyset$ empty, Π largest, 1 identity, dom, ran . $[\alpha]$ power object, $\ni : [\alpha] \rightarrow \alpha$ membership, Λ the power transpose ($f = \Lambda R \iff f \ni = R$), $\tau = \Lambda 1$ singleton, $\bigcup = E \ni$ union, $ER = \Lambda(\ni R)$ existential image. F, G relators; $\text{in} : F(\mu F) \rightarrow \mu F$ the initial algebra, $\langle R \rangle$ its catamorphism; $(\mu X : \varphi X)$ the least solution of $X = \varphi X$. Points: $x R y$.

B&dM dictionary. Everything else on the sheet is the book’s, mirrored through this table and nothing more.

B&dM	here
$S \cdot R$	RS composition reverses
id	1
outl, outr	π, π'
inl, inr	ι, ι'
α	in the initial algebra, not an object
(R)	$\langle R \rangle$
μ	$\bigcup$ (μ is kept for the least fixed point)
$\in, \ni$	$\ni, \ni^\circ$
$T \setminus U, U/T$	$U/T, T \setminus U$ slashes flip, roles kept
$(a, b) \langle R, S \rangle c$	$c \langle R, S \rangle (a, b)$ points reverse too

A row tagged $\checkmark$ has its mirrored statement machine-checked in AOP/A*.lean; the rest are hand-mirrored from the page.

7.1 Minimum and maximum

- 7.1 $\min R \triangleq \exists \cap (\exists^\circ \setminus R) : [\alpha] \rightarrow \alpha$ — a is a minimum of x under R when a is *in* x and is a lower bound of x . R need not be a preorder for the definition to make sense, only for it to be useful.

UNIVERSAL PROPERTY $X \subset \min R \iff X \subset \exists \wedge \exists^\circ X \subset R$.

$$\max R = \min R^\circ.$$

Three consequences of $f(R/S) = (f^\circ S)/R$, all about dividing by $\exists$:

(7.1) $\checkmark \quad \tau(\exists^\circ \setminus R) = R.$

(7.2) $\Lambda(S)(\exists^\circ \setminus R) = S^\circ \setminus R.$

(7.3) $\checkmark \quad \bigcup(\exists^\circ \setminus R) = \exists^\circ \setminus (\exists^\circ \setminus R).$

(7.4) $\checkmark \quad \tau(\min R) = 1 \cap R$ — so R is reflexive exactly when the minimum of a singleton is its inhabitant.

(7.5) $\checkmark \quad \Lambda(S)(\min R) = S \cap (S^\circ \setminus R)$, whose universal property

$$X \subset \Lambda(S)(\min R) \iff X \subset S \wedge S^\circ X \subset R$$

is the workhorse of the chapter; calculations cite it as *universal property of min*.

(7.6) $\checkmark$ **CONTEXT** $\Lambda(S)(\min R) = \Lambda(S)(\min(R \cap S^\circ S))$ — for the purpose of minimising over the sets ΛS returns, R only has to be constrained on values related to one and the same element by S .

Fusion with the power functor

(7.7) $(ES)(\min R) = (\exists S) \cap ((\exists S)^\circ \setminus R)$, from (7.5) and $ES = \Lambda(\exists S)$.

(7.8) Shunting a map through a minimum:
 $(Pf)(\min R) = (\min(fRf^\circ))f.$

(7.9) R reflexive: $(PS)(\min R) = (\exists S) \cap (\exists^\circ \setminus (SR))$. One direction needs tabulations.

(7.10) $\checkmark$ **FUSION WITH THE POWER FUNCTOR**
 $(PS)(\min R) \subset (\exists S) \cap (\exists^\circ \setminus (SR))$, always — this half is all that is used later, and it follows in two lines from the naturality of $\exists$.

Distribution over union

- (7.11) $\checkmark$ R a preorder: $(P(\min R))(\min R) \subset \bigcup(\min R)$ — take a minimum in each set and then a minimum of those. Only an inclusion, because a set in the collection may be empty.
- (7.12) $(P(\min R))(\min R) = (P(\text{dom}(\min R))) \bigcup (\min R)$, using $\min R = (\text{dom}(\min R))(\min R)$.

Implementing min

- 7.1 $\min R$ refines to a map only on non-empty finite sets, and only for R a connected preorder. With $\text{setify} : \text{list}^+ \alpha \rightarrow [\alpha]$ the specification is $\text{minlist } R \subset \text{setify}(\min R)$, and
- $\text{minlist } R = \langle 1, \text{bmin } R \rangle$ with $\text{bmin } R(a, b) = (aRb \rightarrow a, b)$ — the *leftmost* minimum, in the case of ties.

Exercises

- Ex 7.1 $\checkmark$ $\text{subset}^\circ(\ni^\circ \setminus R) = \ni^\circ \setminus R$, where $\text{subset} = \ni/\ni$.
- Ex 7.2 $\checkmark$ $(ER)\text{subset} = (\ni R)/\ni$.
- Ex 7.4 $(PS)(\ni^\circ \setminus R) = \ni^\circ \setminus (SR)$.
- Ex 7.5 $\checkmark$ R a preorder: $(\ni^\circ \setminus R)R = \ni^\circ \setminus R$.
- Ex 7.6 $\checkmark$ $\min(R \cap S) = (\min R) \cap (\min S)$.
- Ex 7.7 $\checkmark$ $\ni^\circ \ni = \Pi$ — every element is in some set; hence $\min R = \ni \iff R = \Pi$.
- Ex 7.8 R, S reflexive: $R \cap S^\circ = (\min S)^\circ(\min R)$.
- Ex 7.9 R reflexive: $R = \ni^\circ(\min R)$.
- Ex 7.10 $\checkmark$ R, S reflexive: $\min R \subset \min S \iff R \subset S$.
- Ex 7.11 $\checkmark$ $\min R$ is simple $\iff R$ is anti-symmetric.
- Ex 7.12 R a preorder: $\min R = \ni \cap ((\min R)R)$.
- Ex 7.13 R a preorder and S a map: $R \cap (SS^\circ)$ is a preorder.
- Ex 7.14 $\checkmark$ R a preorder: $\Lambda(R)(\max R) = R \cap R^\circ$.
- Ex 7.15 $\langle S(\min R), T(\min R) \rangle \subset \Lambda\langle S, T \rangle(\min(R \times R))$.
- Ex 7.16 R reflexive, S a preorder: $\Lambda(\min S)(\min R) = \min(S; R)$, with $S; R = S \cap (S^\circ \Rightarrow R)$.
- Ex 7.19 $\checkmark$ **MINIMAL ELEMENTS** $\text{mnl } R \triangleq \min(R^\circ \Rightarrow R)$.
 $(R^\circ \Rightarrow R)$ is reflexive for every R , but need not be a preorder even when R is.
- Ex 7.20 $\checkmark$ $\min R \subset \text{mnl } R$, with equality exactly when R is a connected preorder.

Ex 7.22	$(Pf)(\text{mnl } R) = (\text{mnl}(fRf^\circ))f$ — Ex 7.8 for mnl .
Ex 7.23	$\text{mnl } R = \ni \iff R$ symmetric.
Ex 7.25	class $Q = \langle 1, \ni \Lambda(Q) \rangle$ cap picks an equivalence class of Q ; for R a preorder, $\text{mnl}(R; S) = \Lambda(\text{mnl } R) \text{ class}(R \cap R^\circ) \text{ mnl } S$.
Ex 7.26	R WELL BOUNDED $\triangleq \text{dom } \ni = \text{dom}(\min R)$ — every non-empty set has a minimum. A well-bounded preorder is necessarily connected, and R is well bounded $\iff$ its strict part $R \cap \neg R^\circ$ is well-founded.
Ex 7.27	R well bounded $\implies fRf^\circ$ well bounded, for any map f .
Ex 7.28	R well bounded $\iff \ni \subset (\min R)R^\circ$.
Ex 7.29	R a well-bounded preorder: $\bigcup(\min R) = (E(\min R))(\min R)$ — (7.11) strengthened to an equality, which is what makes $\text{setify}(\min R)$ a catamorphism.
Ex 7.30	R WELL SUPPORTED $\triangleq \text{dom } \ni = \text{dom}(\text{mnl } R)$ — weaker than well bounded.
Ex 7.31	R a well-supported preorder: $\ni \subset (\text{mnl } R)R^\circ$.
Ex 7.32	R a well-supported preorder: $\bigcup(\text{mnl } R) = (E(\text{mnl } R))(\text{mnl } R)$.

7.2 Monotonic algebras

7.2 ✓ An F -algebra $S : F\alpha \rightarrow \alpha$ is **MONOTONIC ON** $R : \alpha \rightarrow \alpha$ if

$$(FR)S \subset SR.$$

Addition is monotonic on $\leq$: $(\text{leq} \times \text{leq})\text{plus} \subset \text{plus leq}$, i.e. $c = a + b \wedge a \leq a' \wedge b \leq b' \implies c \leq a' + b'$.

For a map: f monotonic on $R \iff f^\circ(FR)f \subset R \iff FR \subset fRf^\circ$; by shunting, f is monotonic on $R \iff$ on R° . Neither equivalence survives for a general S .

f **DISTRIBUTES OVER** R if $F(\min R)f \subset \Lambda((F\exists)f)(\min R)$ — pointwise, for $+$ over $\leq$, $\min x + \min y = \min\{a + b \mid a \in x \wedge b \in y\}$ on non-empty x, y .

T 7.1 ✓ A map f is monotonic over $R \iff$ it distributes over R .

Theorem (Theorem 7.2 — the greedy theorem ✓).

S monotonic on the preorder R° :

$$(\Lambda(S)(\min R)) \subset \Lambda(\llbracket S \rrbracket)(\min R).$$

Take the locally best step at every stage and the result is globally optimal. The proof is the universal property of $\min$, then the hylomorphism theorem: the greedy fold is below every prefixed point of the associated recursion, and transitivity of R discharges it. For $\max$ the condition is monotonicity on R , not R° ; and context can always be brought in, so monotonicity on $R^\circ \cap (\llbracket S \rrbracket^\circ \llbracket S \rrbracket)$ suffices.

Ex 7.33 ✓ $a + b \leq a + b' \implies b \leq b'$, point-free.

Ex 7.34 ✓ in monotonic on $R \implies R$ is reflexive.

Ex 7.37 ✓ A variant: f monotonic on R and $f \subset \Lambda(S)(\min R) \implies \llbracket f \rrbracket \subset \Lambda(\llbracket S \rrbracket)(\min R)$.

Ex 7.38 ✓ S monotonic on $R^\circ \implies (\min(FR))\Lambda(S)(\min R) \subset (ES)(\min R)$ — optimal substeps.

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- Ex 7.39 $\text{takewhile } p = \Lambda((\text{list } p)(\text{prefix}))(\text{max } R)$ with $R = \text{length leq length}^\circ$; with $\text{prefix} = \langle \text{nil}, \text{cons} \cup \text{nil} \rangle$ the greedy theorem gives the standard program.
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- Ex 7.40 **MAXIMUM SEGMENT SUM** $\text{mss} = \Lambda(\text{segment sum})(\text{max})$, and with $\text{segment} = \text{suffix prefix}$, $\text{mss} = \Lambda \text{ suffix } P(\Lambda(\text{prefix sum})(\text{max}))(\text{max})$; the greedy theorem gives $\langle \text{zero}, \text{oplus} \rangle \subset \Lambda(\text{prefix sum})(\text{max})$ with $\text{oplus} = \Lambda(\text{zero} \cup \text{plus})(\text{max})$, and a linear-time program follows.
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- Ex 7.41 $\text{filter } p = \Lambda((\text{list } p)(\text{subseq}))(\text{max } R)$; with $\text{subseq} = \langle \text{nil}, \text{cons} \cup \pi' \rangle$ the greedy theorem gives the standard program.
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- Ex 7.42 L the lexical order: $(1 \times L)\text{cons} \subset \text{cons } L$, hence $\langle \text{nil}, \Lambda(\text{cons} \cup \pi')(\text{max } L) \rangle \subset \Lambda(\text{subseq})(\text{max } L)$; the greedy step is $(\text{ok} \rightarrow \text{cons}, \pi')$ with $\text{ok}(a, x)$ iff $x = []$ or $a \geq \text{head } x$.
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7.3 Planning a company party

7.3 ✓ A company is a tree $\alpha ::= \text{node}(\alpha, \text{list}(\text{tree } \alpha))$; each employee has a *conviviality* rating, and no employee may attend with their immediate supervisor. Maximise the total rating of the guest list. Base functor $F(A, B) = A \times \text{list } B$.

$\Lambda \text{ party}(\max R)$ with $R = (\text{list rating sum}) \text{ leq } (\text{list rating sum})^\circ$ and $\text{party} = \llbracket \langle \text{include}, \text{exclude} \rangle \rrbracket \text{ choose}$, $\text{choose} = \pi \cup \pi'$ — one catamorphism producing the two parties, one with the root and one without.

$\text{include} = (1 \times (\text{list } \pi' \text{ concat})) \text{ cons}$ (a map: take the root, so the roots of the immediate subtrees are excluded), $\text{exclude} = (1 \times (\text{list choose concat})) \pi'$ (not a map: each immediate subtree is free to include or exclude its root).

$\Lambda \text{ party}(\max R)$
 $=$ definition of party
 $\Lambda(\llbracket \langle \text{include}, \text{exclude} \rangle \rrbracket \text{ choose})(\max R)$
 $=$ $\Lambda(XY) = \Lambda(X)(EY)$
 $\Lambda \llbracket \langle \text{include}, \text{exclude} \rangle \rrbracket (E \text{ choose})(\max R)$
 $\supseteq$ Ex 7.38, since choose is monotonic on R
 $\Lambda \llbracket \langle \text{include}, \text{exclude} \rangle \rrbracket (\max(R \times R)) \Lambda \text{ choose}(\max R)$
 $\supseteq$ greedy, since $\langle \text{include}, \text{exclude} \rangle$ is monotonic on $R \times R$
 $\llbracket \Lambda \langle \text{include}, \text{exclude} \rangle (\max(R \times R)) \rrbracket \Lambda \text{ choose}(\max R)$

7.3 ✓ The two conditions are $(R \times R) \text{ choose} \subset \text{choose } R$ and $(1 \times \text{list}(R \times R)) \langle \text{include}, \text{exclude} \rangle \subset \langle \text{include}, \text{exclude} \rangle (R \times R)$, the second reducing by Ex 7.15 to $\Lambda \text{ include}(\max R)$ and $\Lambda \text{ exclude}(\max R)$ separately.

include is already a map, and exclude refines to $\pi'(\text{list}(\Lambda \text{ choose}(\max R))) \text{ concat}$. The result: a linear-time greedy sweep for a problem posed as an exercise in dynamic programming.

7.4 Shortest paths on a cylinder

7.4 ✓ An $n \times m$ array of positive integers rolled into a cylinder, the top and bottom rows adjacent. A path enters at the left, and from a square may step to one of the three adjacent squares of the next column. Find a path of least total cost, in $O(nm)$.

N is the type functor of n -tuples, $L = \text{list}^+$ with base functor $F(A, X) = A + (A \times X)$. The problem is $\Lambda \text{ paths}(\min R)$ with $R = \text{sum leq sum}^\circ$ and $\text{paths} : L \ N \ \text{Nat} \rightarrow [\alpha]$.

paths is not the transpose of a catamorphism, so it is built from one: $\text{paths} = \llbracket \text{generate} \rrbracket \text{setify} \bigcup$ with $\text{generate} : F(N \ \alpha, N \ P \ L \ \alpha) \rightarrow N \ P \ L \ \alpha$, a *lax natural transformation*, assembled by type from $\text{generate} = F(1, \text{moves trans } N \bigcup) \text{zip cp } N(P \text{ in})$.

$\text{moves } x = \{\text{up } x, x, \text{down } x\}$ rotates columns, trans transposes a set of tuples, zip commutes N with F , $\text{cp} = \Lambda(F(1, \ni))$.

(7.13) ✓ The crux is Theorem 7.1, not the greedy theorem: in is a monotonic map, hence distributes, so

$$F(1, \min R) \text{ in } \subset \text{cp } (P \text{ in}) (\min R).$$

$\Lambda \text{ paths} (\min R)$

= definition of paths

$\Lambda \llbracket \text{generate} \rrbracket \text{setify} \bigcup (\min R)$

$\supseteq$ distribution over union (7.11), R a preorder

$\Lambda \llbracket \text{generate} \rrbracket \text{setify} (P(\min R))(\min R)$

$\supseteq$ naturality of setify

$\Lambda \llbracket \text{generate} \rrbracket (N(\min R)) \text{setify} (\min R)$

$\supseteq$ fusion, for the Q below

$\llbracket Q \rrbracket \text{setify} (\min R)$

7.4 ✓ The fusion condition is $\text{generate } N(\min R) \supset F(1, N(\min R))Q$, and unfolding it through (7.13) and the naturalities of zip , trans and moves gives $Q = \text{moves trans } (N \ P(\min R))(N(\min R)) F(1, \cdot) \text{zip } N \text{ in}$, which as a coproduct is $Q = [N \ \text{wrap}, (1 \times (\text{moves trans } N(\min R))) \text{zip}' N \ \text{cons}]$.

Sets and n -tuples both become lists, $\min R$ becomes $\text{minlist } R$, and zip the standard function — a single left-to-right sweep, one column at a time.

7.5 The security van problem

- 7.5 ✓ A bank's cash must stay between 0 and N ; a security van tops it up or takes a surplus. Given a sequence of transactions, cut it into as few *secure* segments as possible. The moral of the section: when the monotonicity condition fails, refine the *ordering* instead of abandoning the greedy algorithm.

ceiling = $\Lambda(\text{prefix sum})(\text{max leq})$ and floor = $\Lambda(\text{prefix sum})(\text{min leq})$; x is secure iff $\text{bmax}(\text{ceiling } x, \text{ceiling } x - \text{floor } x) \leq N$. secure is the corresponding coreflexive.

The problem is $\Lambda(\text{partition list secure})(\text{min } R)$ with $R = \text{length leq length}^\circ$; fusing list secure into $\text{partition} = \langle \text{nil}, \text{new} \cup \text{glue} \rangle$ gives

$$\text{partition list secure} = \langle \text{nil}, \text{new} \cup \text{old} \rangle,$$

$$\text{old} = (1 \times \text{cons}^\circ) \text{assocl}((\text{cons secure}) \times 1) \text{cons}.$$

Greedy needs $[\text{nil}, \text{new} \cup \text{old}]$ monotonic on R° , i.e.

$$(7.14) \quad (1 \times R^\circ) \text{new} \subset (\text{new} \cup \text{old}) R^\circ \text{ — true.}$$

$$(7.15) \quad (1 \times R^\circ) \text{old} \subset (\text{new} \cup \text{old}) R^\circ \text{ — false: two partitions of the same sequence may have equal length without } \text{secure}([a] \# x) \text{ implying } \text{secure}([a] \# y).$$

secure is *prefix-closed* — secure prefix $\subset$ prefix secure — so refine R to $R; H$, where $H = (\text{head prefix head}^\circ) \cup (\text{nil}^\circ \text{ nil})$ and $R; H = R \cap (R^\circ \Rightarrow H)$: same length, and the first component a prefix.

With the refined order, (7.16) $(1 \times (R; H)^\circ) \text{new} \subset (R; H)^\circ (\text{new} \cup \text{old})$ follows from (7.18) $(1 \times \Pi) \text{new} \subset \text{new } H^\circ$, and (7.17) for old from the three claims (7.19)–(7.21) about the strict part $|R| = R \cap \neg R^\circ$. Hence

$$\langle (\text{ok} \rightarrow \text{glue}, \text{new}) (\text{wrap wrap}) \rangle \subset \langle \Lambda(S)(\text{min}(R; H)) \rangle,$$

with ok holding at (a, xs) when $xs \neq []$ and $[a] \# \text{head } xs$ is secure.

secure turns out to be suffix-closed too, which is what makes the test efficient: ceiling = $\langle \text{zero}, \text{omax plus} \rangle$ and floor = $\langle \text{zero}, \text{omin plus} \rangle$ (two baby applications of the greedy theorem), so

ok $N(a, [x] \uplus xs) = (\text{omax}(a + \text{ceiling } x) - \text{omin}(a + \text{floor } x) \leq N)$.

Ex 7.57 The refined order can be given directly: $Q = (\text{nil}^\circ!) \cup (\text{prefix}; (\text{tail}^\circ Q \text{tail}))$, a preorder and a linear order on partitions of one sequence. $Q \not\subseteq R$, yet $\Lambda(\llbracket S \rrbracket)(\min Q) \subset \Lambda(\llbracket S \rrbracket)(\min R)$ whenever secure is suffix-closed.